

EXPERIMENT: UNIFICATION AND RESOLUTION – EXERCISES WITH SOLUTIONS

AIM

To understand and implement unification and resolution concepts in PROLOG using facts, compound terms, rules, and queries.

OBSERVATION

1. Sentence Matcher

Problem: Match sentences stored as facts and retrieve all possible values.

Program:

```
% Facts
sentence(the_cat_sits).
sentence(the_dog_runs).

% Query
% ?- sentence(X).
```

Output:

```
X = the_cat_sits ;
X = the_dog_runs ;
false.
```

Observation:

The query matches the variable X with each sentence stored in the knowledge base. PROLOG returns all possible solutions using backtracking.

2. Compound Term Unification

Problem: Unify a compound term and determine the value of a variable.

Program:

```
% Facts
likes(john, food(pizza, spicy)).
likes(mary, food(pizza, sweet)).

% Query
% ?- likes(john, food(pizza, T)).
```

Output:

T = spicy.

Observation:

PROLOG successfully unifies the query with the matching fact. The variable T is bound to spicy because the compound terms food(pizza, T) and food(pizza, spicy) match.

3. Conflict Resolution

Problem: Determine whether two rules support the same fact.

Program:

```
% Facts
rule1(climate_change).
rule2(climate_change).

% Rule
conflict(X) :- rule1(X), rule2(X).

% Query
% ?- conflict(climate_change).
```

Output:

true.

Observation:

The query succeeds because both `rule1(climate_change)` and `rule2(climate_change)` are true. PROLOG resolves the query by satisfying both conditions in the rule.

RESULT

Thus, the concepts of unification and resolution were successfully demonstrated using sentence matching, compound term unification, and conflict resolution in PROLOG.